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Review

A Review of the Development and Future Challenges of Case-Based Reasoning

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Abstract: Case-based reasoning (CBR), which is based on the cognitive assumption that similar problems have similar solutions, is an important problem-solving and learning method in the field of artificial intelligence (AI). In this article, the development of CBR is reviewed, and the major challenges of CBR are summarized. The paper is organized into four parts. First, the basic framework and concepts of CBR are introduced. Then, the developed technology and innovative work that were designed to solve problems by CBR are summarized. Then, the application fields of CBR are summarized. Finally, according to the idea of deep learning and interpretable AI, the main challenges for the future development of CBR are proposed.

Keywords: case-based reasoning; problem solving; similarity measure; case retrieval; case adaptation; case base maintenance; deep learning

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1. Introduction

Case-based reasoning (CBR), which belongs to a branch of knowledge-based systems, is an important research direction in the field of artificial intelligence (AI) [1]. It is a new reasoning method developed along with the research of cognitive science. CBR simulates the cognitive process of human beings. The core idea is to use the solutions of similar problems from past cases to reason and solve new problems based on a cognitive hypothesis: similar problems have similar solutions [2]. Since CBR was proposed, it has gradually formed a developed reasoning model framework and has become an effective and practical AI technology that is widely used in industrial control [3], emergency decision making [4], planning and design [5], medical diagnosis [6], and other fields. We select the EI database to explore the growth trend of CBR literature, which is the most extensive and complete engineering literature database in the world, covering many fields and subjects. Search for 'case-based reasoning (or case based reasoning)' in the option column of 'Subject/Title/Abstract', and count the literature number related to CBR every 5 years. The changing trend from 1992 to 2020 is shown in Figure 1. It can be seen that the scale and quantity of literature on CBR research and application are gradually increasing.

CBR first appeared in the description of 'Dynamic Memory' by Schank and Abelson [7] of Yale University, which laid a theoretical foundation for the generation of CBR. Then, Riesbeck and Schank [8] described the classic definition of CBR: CBR solves problems by using or adjusting solutions to old problems. It is a problem-solving paradigm that is fundamentally different from other major AI approaches [9]. Instead of relying solely on the general knowledge of a problem domain or making associations based on generalized relationships between problem descriptors and conclusions, CBR can use the specific knowledge of specific problem situations (cases) experienced in the past to solve similar, new problems. A second important difference is that CBR is an incremental, continuous learning method that retains new experiences in solving each problem and then applies it to

solve new problems in the future. Aamodt and Plaza [9] summarized the reasoning process of CBR into the following four steps: Retrieve, Re-use, Revise, and Retain; this process is called the '4R' cycle, which provides a basic reasoning framework for CBR research. In 1999, Watson [10] proposed that CBR is a methodology rather than a technology, which means that CBR can use various technologies to achieve the solution to a problem, interpretation, and learning process. Thus, CBR can continue to develop as researchers are faced with the challenge of applying it to various technologies. After Watson, Finnie and Sun [11] integrated the case representation into '4R' and proposed the '5R' model of CBR, which improved the reasoning framework of CBR.

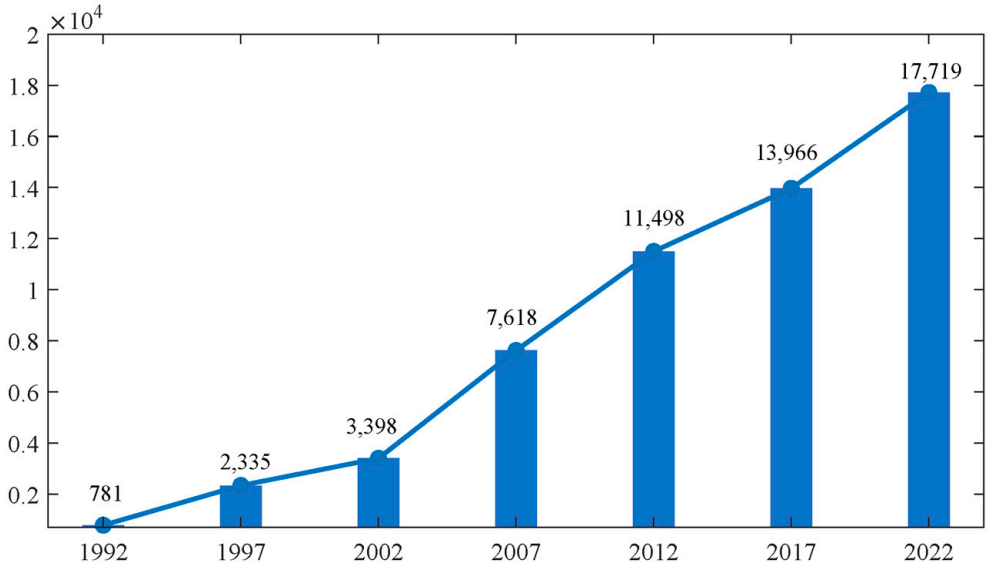


Figure 1. The changing trend of quantity of literature around CBR.

As a mature research field, CBR has many specific methods in each step, wide application fields, and broad development prospects; however, this also makes it difficult to have a comprehensive understanding of this field. Therefore, this paper provides a detailed analysis and summary of various aspects of CBR. First, it classifies and summarizes the concepts, frameworks, and key technologies of each step of CBR, providing a comparison of the advantages and disadvantages of CBR methods, and summarizing the research direction of algorithm development. Second, it summarizes the specific applications of CBR in different fields and provides relevant literature, showing the current application status of CBR in each field. Finally, according to the research status of CBR and the current development frontier of AI technology, we propose the two possible development directions of CBR, looking forward to the future research trends of CBR. It is hoped that this paper can provide useful help for beginners, practitioners, and researchers in this field.

The rest of this paper is organized as follows: in Section 2, the basic framework and basic concepts of CBR are introduced; in Section 3, the development of key technologies of CBR is reviewed; in Section 4, the application fields of CBR are discussed; finally, in Section 5, several challenges regarding the future development of CBR are presented.

2. Basic Framework and Concept of CBR

The CBR framework [9] is shown in Figure 2. The main functions of each step are as follows:

(1) Case retrieval: One or more source cases most similar to the new case are retrieved from the case base.
(2) Case re-use: Information and knowledge from similar cases are re-used to establish solutions adapted to new case.
(3) Case revision: The proposed solution is evaluated, and the solution is adjusted if it does not meet the requirements.

(4) Case retention: The parts of this experience that may be useful for solving problems in the future are retained.

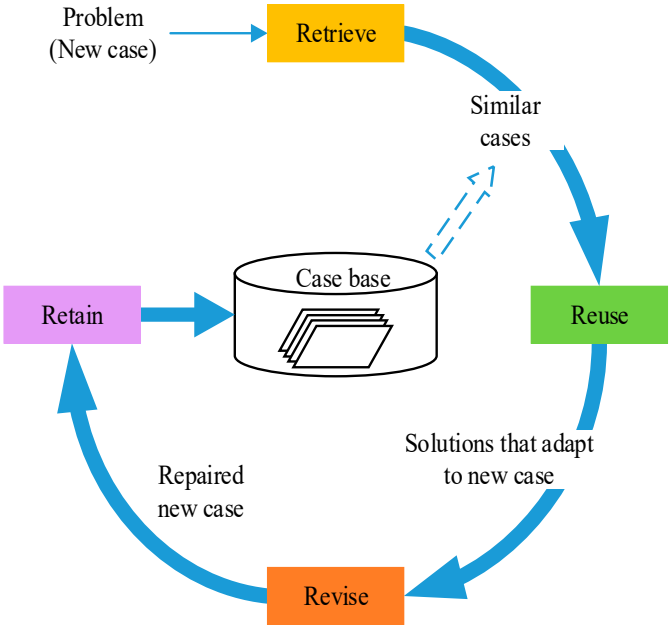


Figure 2. CBR cyclic structure.

From the problem-solving process of CBR, it can be seen that the initial description of the problem will be defined as a new case. Then, the similarity measurement between the new case and the source case in the case base is carried out to achieve case retrieval. Therefore, the case is the basis of CBR, and the representation form of the case plays an important role in CBR. The solutions for similar cases are then directly and appropriately adapted as solutions for new cases through the retrieval and re-use steps, and new solutions are evaluated to determine whether revision is needed. Finally, whether a new case and solution need to be retained for the solution of subsequent problems is considered. Several key links affecting the reasoning performance are case representation, case retrieval, case re-use, case retention, and case base maintenance.

It can be seen from the recent collected papers from the International Conference on Case-Based Reasoning (ICCBR) that articles on similarity measurement, retrieval, adaptation, and other methods have increased, which indicates that research on CBR key technologies is still very important to the CBR community. According to the basic concept of CBR and the principle of problem solving, we retrieve some keywords in the theme, title and abstract of the literature based on CBR in the EI database, which includes 'case representation', 'similarity', 'case retrieval' or 'retrieve', 'case adaptation' and 'case revision', and 'case base maintenance' or 'case maintenance'. Literature retrieval was carried out from these dimensions. The number of relevant literatures is shown in Figure 3. Figure 3 shows that case representation, similarity measurement, case retrieval, case adaptation, and case base maintenance have received sufficient attention. The following will review and summarize these dimensions.

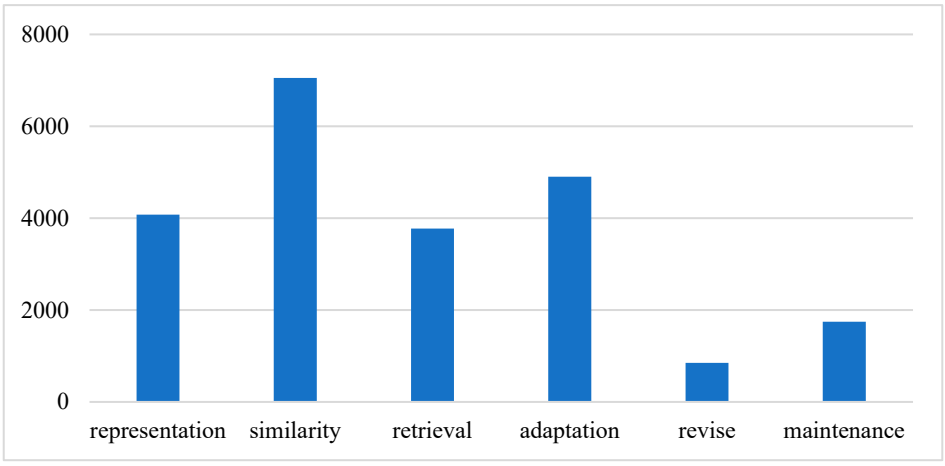


Figure 3. The quantity of CBR key technology research literature from 1992 to 2022. In the EI database, keywords are searched in the subject, title, and abstract based on CBR literature: case representation, similarity, retrieval, case adaptation, case revision, and case base maintenance or case maintenance.

3. Development of CBR Key Technologies

Research on the key technologies of CBR promotes the rapid development of CBR. The development of these key technologies is inseparable from the assistance of other intelligent algorithms, as shown in Figure 4. In this section, the research status and development trends of key CBR technologies, including case representation, similarity measurement, case retrieval, case adaptation, and case base maintenance, are discussed.

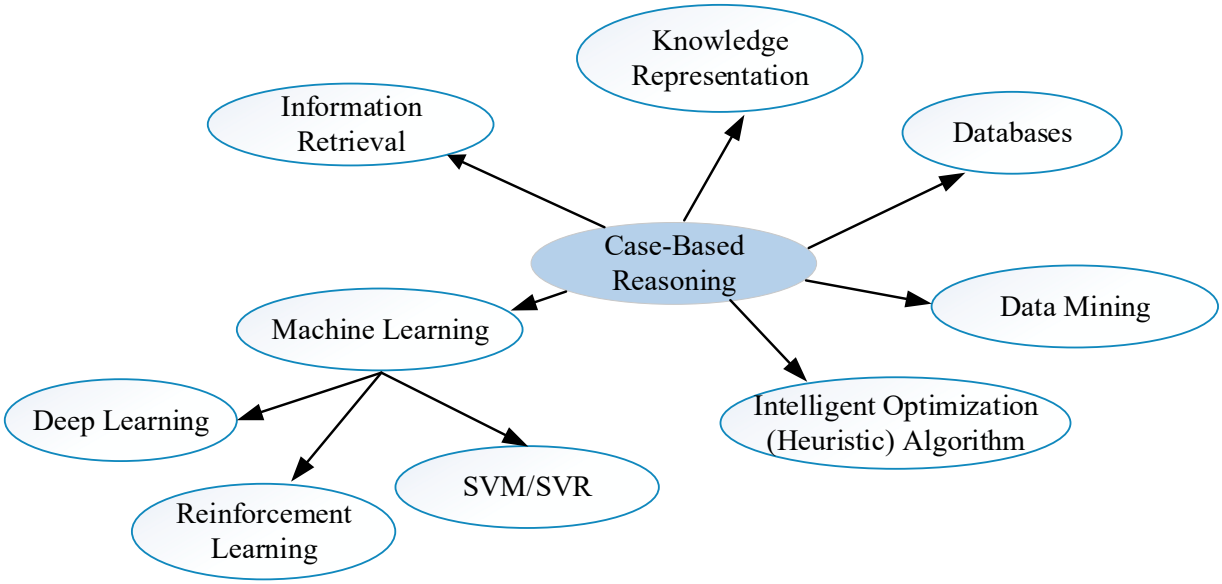


Figure 4. CBR and other AI technologies.

3.1. Case Representation

The case is a knowledge representation of experience [12], including the content of past lessons learned and the context in which these lessons can be used. Bergmann et al. [13] mentioned the case as a contextualized piece of knowledge representing an experience that teaches a lesson fundamental to achieving the goals of the reasoner. The case representation focuses on what is stored in the case base and how to build the case to describe its contents [9]. Generally, it can be expressed as a binary group:

$$case = \{P, S\} \quad (1)$$

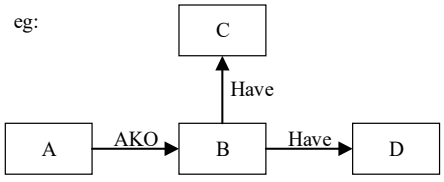
where P and S refer to a set of features describing the problem and a set of features describing the solution, respectively. In addition, if it is necessary to know the result of the solution, the case is represented as a triplet [14]:

$$case = \{P, S, O\} \tag{2}$$

where O represents a set of features describing the result.

Problem descriptions describe the relevant information about a case in the form of attributes or features and in the form of images and sequences, etc. [15]. Case representation methods mainly include feature vector [16], frame representation [17], object-oriented case representation [18], predicate-based representation [19], semantic nets [20], and production rule representation [21], etc. Generally, the appropriate representation method is selected according to the different application fields. The specific description is shown in Table 1.

Table 1. Case representation methods.

Method	Element	Representation	Limitation
Frame	slot: facet: value	<Frame name> slot 1: facet 1_1 value 1_{11}, value 1_{12}, ⋯ ⋮ facet 1_m value 1_{m1}, value 1_{m2}, ⋯ ⋮ slot n: facet n_1 value n_{11}, value n_{12}, ⋯ ⋮ facet n_m value n_{m1}, value n_{m2}, ⋯ constraint: constraint condition	Low reasoning efficiency; Hard to track and control.
Object-Oriented	CLASS::=<ID, DS, MS, MI> ID: Identifier DS: Data Structure MS: Method Set MI: Message Interface	class <name>[:<Superclass>] [<Class variable name>] Structure <Static structure description of an object> Method <Definition of an operation on an object> Restraint <Restricted condition> END	
Production Rule	<production>::= <precondition> <conclusion>	P→Q IF P THEN Q (CF = [0, 1]) CF: Certainty Factor	Low efficiency; Unable to express structured knowledge
Semantic Nets	(Node1, Arc, Node2) Semantic Relation	eg:  AKO: A-Kind-Of	Non-rigidity; Low reasoning efficiency; Knowledge access complexity
Predicate-Based	Predicate (Constant/Variate/Function) Conjunctions Quantifier	Predicate Formula	Cannot represent uncertain knowledge; Combinatorial explosion; Low efficiency

In recent years, with the continuous expansion of the application field of CBR, the complexity of many cases under specific problems is deepening, resulting in the diversification and increased complexity of case representation. Examples include implicit, empirical, and unstructured design knowledge in complex product design [22] as well as case text

representation in medical information [23]. At the same time, in the era of big data, the massive flood of information has led to a surge in the number of cases; consequently, the description of case information is messy, and the scale of the case base is expanding [24]. In the face of this situation, these mature case representation methods are more applicable to cases with simple structures, single representation methods, and fewer parameters, and cannot be applied well to complex and changeable cases. Therefore, many scholars have begun to study new methods to solve the above problems.

On the one hand, based on the existing case representation method, new improvement strategies have been proposed to achieve the complex representation of cases in combination with other AI methods according to the actual application. For example, a framework case representation method integrating multiple information enhances the representability of the case [25]. A multi-strategy ontology mapping method can be applied to realize the semantic expression between the process knowledge graph and the entity model [26]. An association representation between cases is added to case representation [27]. However, the problem is that improved methods are often aimed at a specific application field, which requires domain-specific knowledge in order to build a model. This method is complex and only applicable in a small range of cases. Second, it increases the workload of the case representation process. For ordinary users, it is often difficult to manipulate the model. Finally, there are many improvement methods, but explanations of their advantages and disadvantages are lacking. Therefore, choosing different AI methods is a problem that must be considered.

On the other hand, the CBR community is actively providing new representations of cases through deep learning (DL) technology and other forms of vectorization. For example, a pre-trained word2vec model and a one-dimensional convolutional neural network were used to generate appropriate case representations [28], and a deep autoencoder and Siamese neural network (SNN) were used to generate text case representations [29]. This type of exploration provides corresponding processing methods for complex elements such as text and images in each case, but supervised learning based on DL has problems such as data labeling, hyperparameter optimization, network structure, and parameter design. Furthermore, how to express the rich structured cases of CBR in a network context is another key issue [30]. Finally, determining whether each new case generation and preservation need to update the network also requires relevant discussion and research.

3.2. Similarity Measure and Case Retrieval

An important step in CBR is to retrieve similar cases. Case retrieval depends on an accurate assessment of its similarity to the target problem. This process involves the following two key factors: (1) Similarity measure, where the similarity between cases is an important basis for retrieving similar cases; (2) Retrieval method, where choosing the appropriate retrieval method can improve retrieval speed and accuracy.

3.2.1. Similarity Measure

In CBR, similarity is a bridge between case representation and retrieval. Aamodt and Plaza [9] thoroughly introduced similarity evaluations, which are mainly divided into surface similarity, structural similarity, and a similarity framework. This method is mostly selected according to the case representation method; for example, feature representation often uses surface similarity calculation, and object-oriented case representation can use structural similarity calculation [31]. There are two mathematical methods for representing similarity: relationship description and function representation. Table 2 provides the details regarding similarity.

Table 2. The relation representation and function representation of similarity.

Relation			Function		
$SIM(x,y)$	$\Leftrightarrow$	x and y are similar	$SIM(x,y)=1$	$\Leftrightarrow$	x and y are exactly similar
$DSIM(x,y)$	$\Leftrightarrow$	x and y are dissimilar	$SIM(x,y)=0$	$\Leftrightarrow$	x and y are exactly dissimilar
$R(x,y,z)$	$\Leftrightarrow$	x is at least as similar to y as x to z	$0<SIM(x,y)<1$	$\Leftrightarrow$	x and y are partly similar

The most commonly used method is to calculate the similarity between cases x and y by a distance metric function (taking Euclidean distance as an example):

$$SIM(x,y)=1-DIST(x,y)=1-\sqrt{\sum_i w_i^2 dist^2(x_i,y_i)} \quad (3)$$

where x_i and y_i represent the same feature of the two cases, w_i is the corresponding weight, and $SIM(x,y)\in[0,1]$.

With the continuous expansion of CBR applications, there have been difficulties in obtaining similarity measurements based on complex or mixed data such as time series, images (or graph structure), or text in complex CBR tasks. However, since the metric function calculation is mainly applicable to the case of numerical attribute descriptions, it cannot be directly applied to cases containing multiple complex representations; therefore, improving retrieval performance by developing more effective similarity measurement methods has become the focus of research in the field of CBR, such as [32–34]. At present, many scholars have improved the measurement algorithm in the following three directions:

(1) An improved method based on a hybrid similarity measure [32,35–37] mainly improves the calculation accuracy of similarity by processing attribute features, such as adding other information and setting multiple attribute value formats. This hybrid measurement method solves the similarity measurement problem of multi-attribute representation cases, but it often needs to combine the relevant knowledge of specific applications, which is a highly professional task and computationally complex.

(2) Based on the weighted similarity measure of feature weight optimization [38–42], the measurement calculation is improved through the reasonable distribution of feature weight (i.e., w_i in Formula (3)). The emphasis is on the selection, optimization, and improvement of the weight distribution method. This kind of method has been studied for the longest amount of time, and the achievements are more fruitful. It combines information entropy, genetic algorithm, neural network, and other optimization algorithms. However, due to the different evaluation criteria in each article, how to choose the appropriate optimization algorithm in practical applications is a difficult problem.

(3) Based on the (deep) metric learning algorithm [33,43,44], the learning of the similarity measure is achieved by training a (deep) neural network. This method has a shorter research time in CBR compared with that of other case similarity measurement algorithms. Its advantage is that it realizes similarity calculations in the form of a neural network, solves the nonlinear problem, and reduces the computational complexity of the similarity measurement process. In dealing with cases represented by text and images, etc., the network structure has better representation advantages in case data processing; however, problems such as the neural network design caused by this method also need to be considered and studied.

In addition, there are some other intelligent methods combined with similarity measurement methods to improve the retrieval performance [45–48]. However, due to the large number of AI methods, this article only lists several articles that the author believes have been more valuable in recent years for reference. Verma et al. [45] proposed a data-driven method to model the local similarity measure of numerical and class attributes. Lenz et al. [46] studied an ontology-based semantic similarity measure in the application of argumentation schemes. Zeyen et al. [47] focused on the similarity measures of semantic label graphs and proposed a combination of A* search and knowledge-intensive local simi-

larity measures, which not only outputs the similarity but also provides the corresponding mapping that can be used for interpretation and adjustment.

3.2.2. Case Retrieval

As an important step of case-based reasoning, obtaining the correct case retrieval results determines the overall performance of CBR systems. Effective retrieval means not only finding similar cases but also finding useful similar cases. To date, many mature retrieval methods have been formed, including the nearest neighbor algorithm (NN) [49], knowledge-guided approach [50], template retrieval [51], two-level retrieval [52], and index-based retrieval [53], etc. The efficiency of these retrieval methods depends to a large extent on the following:

- The representation of the objects.
- The case base structure.
- The similarity measure.
- The accuracy of the intended answer or solution.

In case retrieval, similarity-based retrieval methods [35,36,54] are the most interesting to scholars and have been widely used. The classic representative is NN (KNN), which is the most commonly used method in case retrieval because its principle is easy to understand and simple to calculate. However, for complex large-scale data, the simplicity of the method also limits its retrieval in complex cases; therefore, NN is often combined with other algorithms to improve retrieval accuracy and efficiency [55–57]. Although purely similarity-based retrieval is still the most widely used technology, the limitations of similarity are gradually exposed with the development of CBR; for example, the results obtained by retrieval are singular and cannot provide users with more novel and valuable references. There may be no similar cases or only a few cases with similar features in the case base; this leads to a no-retrieval result. Alternatively, the retrieval results cannot provide support for subsequent re-use stages; therefore, while similarity continues to play a role in retrieval, it also gradually combines with other standards to guide the retrieval process.

(1) Retrieval based on diversity is often used in recommendation systems to provide users with more diversified solutions to avoid similar recommended single and limited cases [58,59].
(2) Retrieval oriented to adaptation guidance [60] is based on the assumption that the most similar situation may not be adapted and can perform preliminary adaptation work during the retrieval process, which can significantly reduce adaptation failure and adaptation costs [61,62].
(3) Interpretation-oriented retrieval, which explains CBR and justifies recommendations or solutions, is often important, especially in the field of medical decision making, in which explaining the cause and correctness of the search results can provide compelling support [63–65].

In addition, selecting different retrieval methods according to case representation is also a key research direction for many scholars; for example, text language retrieval combined with natural language processing (NLP) technology [66,67], retrieval based on non-symbolic types such as images [68], or case retrieval with missing case information [69]. These methods mainly aim at the analysis of the features of the case data itself, combining feature selection and measurement algorithms to design the case retrieval method.

3.3. Case Adaptation

The re-use phase needs to use the experience and knowledge of old cases in new situations and obtain the solution to the new problem by adapting the retrieval results. Figure 5 shows the general re-use principle of a selected case. The general process of this phase is shown in Figure 5. If the new problem is the same as the old case retrieved, then the re-use phase only needs to copy the old case solution; however, in reality, the situation with new problems is often not exactly the same as that of old cases, and the retrieved

solution can only be regarded as an initial reference solution. Any difference between the new problem and the retrieved case may require appropriate adaptation of the solution to adapt to the new problem. Especially when CBR is used for constructive problems, such as design tasks, planning tasks, and decision-making tasks, adaptation is often essential. Because such tasks have difficulty finding every solution in the case base, the adaptation process is essential.

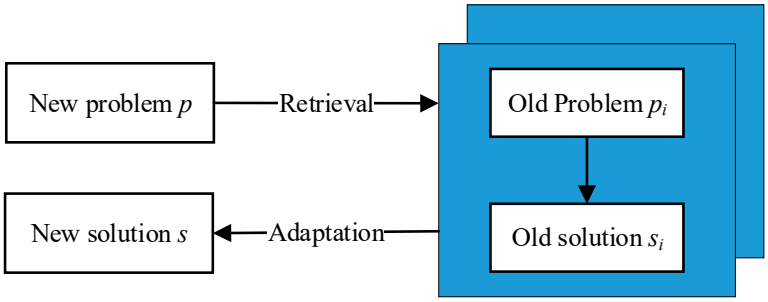


Figure 5. Re-use principle.

The complexity of the adaptation method is reflected in two dimensions: what is changed in the retrieved solution and how the change is achieved [15]. The basic adaptation type is shown in Figure 6, of which the following three types truly need adaptation operations:

- Substitutions: They replace some part of the retrieved solution by another or by several others.
- Structural transformations: They alter the structure of the solution and re-organize the solution by adding, deleting, or replacing parts of the proposed solution [70].
- Generative adaptations: They replay the method of deriving the retrieved solution on the new problem. This is the most complex form of adaptation.

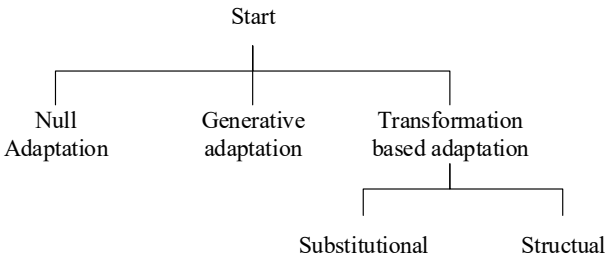


Figure 6. Basic adaptation types.

The most important part of the adaptation process is the learning of adaptation knowledge, which is also a classic problem in CBR. In the early stage, CBR invested much energy in developing case adaptation knowledge, but the difficulty of knowledge generation seriously hindered the development of CBR. It can also be seen from Figure 3 that there is little previous literature on adaptation. Later in the research process, the potential of learning methods to acquire case adaptation knowledge was gradually recognized, including generating rules through decision tree learning [71,72] and support vector regression (SVR) [51,73], and by applying CBR to the adaptation process itself [74,75]. The case-difference heuristic (CDH) first proposed by Hanney and Keane [76] has become one of the most commonly used methods for learning adaptation knowledge. Equation (4) describes the difference between case x and case y:

$$\triangle(x, y) = (x_1 - y_1, \ldots, x_k - y_k, s_x - s_y) = (\triangle f_1, \ldots, \triangle f_k, \triangle s) \quad (4)$$

where s represents the solution for the case, $\triangle f_i$ represents the i-th feature difference, and $\triangle s$ represents the scheme difference. CDH is used to attribute the difference in the solution

description to the difference in its problem description, transforming the difference into rules, thus achieving the case adaptation process.

At present, based on the complexity of case knowledge, methods of learning and adapting knowledge are mainly studied from the following directions:

(1) Improvement of the CDH method [61,77–79]. Although the CDH adaptation method reduces the burden of knowledge engineering, it also brings the problem of defining the difference function. Recent studies have used implicit calculation in ML technology to replace the traditional CDH difference calculation, but there is still a lack of theoretical proof.
(2) Adaptation method based on knowledge and rules [80–82]. Lieber et al. [81] proposed using positive examples to adapt rules, while using negative examples to filter out some of the rules to avoid the problem of incorrect schemes in CBR systems during the adaptation process. These methods focus on the extraction efficiency of knowledge and rules to enhance adaptation performance.
(3) Adaptation methods based on machine learning (ML) or DL [51,83,84]. Long et al. [84] proposed a feature re-use case adaptation (FR-CA) method based on an SVR machine, which can automatically and intelligently solve product experience features and achieve the integration of expert comprehensive decisions with the least expert participation. These methods simplify the compilation and extraction of adapted knowledge through an end-to-end learning process, but the design of network structures often affects the learning effect.

These learning-based case adaptation methods often use different evaluation criteria to evaluate the performance of the adaptation model in different tasks. According to task requirements such as classification and diagnosis tasks, accuracy, F1 score, and recall rate are often used as evaluation criteria for the model. In tasks such as timing or demand forecasting, various error formulas and R^2 are used as evaluation criteria. In addition, the next step in case adaptation in CBR is case revise, i.e., a further evaluation of the adaptation results; however, this step often needs to be implemented after the model is actually applied to practice.

3.4. Case Base Maintenance (CBM)

As CBR systems were developed and deployed for real-world application scenarios, the potential pitfalls of long-term case learning became apparent, especially in relation to the impact of case-based growth on retrieval costs [15]. In the process of case studies, cases are continuously saved to the case base (CB). On the one hand, the increase in the number of cases helps to improve the problem-solving ability of CBR systems. However, with the increase in the CB scale, redundant, repetitive, or even wrong cases increase the time complexity of the systems and reduce the overall performance of the CBR system. Especially in the current era of information flooding, many case bases are already very large premises. Case base maintenance (CBM) has also begun to become more important, and related research is gradually increasing.

CBM ensures the performance of CBR systems by deleting, adding, updating, and modifying cases and related data in the case base. Leake and Wilson [85] defined CBM as the method used in CBR to operate and organize the content of its case base in order to improve or maintain the ability of CBR technology. We summarize the general CBM maintenance process in Figure 7. The basic requirement of CBM is that CBR systems need to maintain the existing case base without affecting its problem-solving capability [86]. Smiti et al. [87] first proposed dividing CBM into the following two strategies: case base division and case base optimization. In recent years, Nakhjiri et al. [88] proposed dividing CBM into the following steps:

(1) Direct models [89–91]: They do not consider the relationship between cases and do not otherwise retrieve information from individual cases.
(2) Hybrid model [87,92]: They describe an association between cases by integrating different AI methods.

(3) Case property model [93,94]: They integrate additional information into the case to better illustrate the features of the case in the case base to achieve the maintenance function.

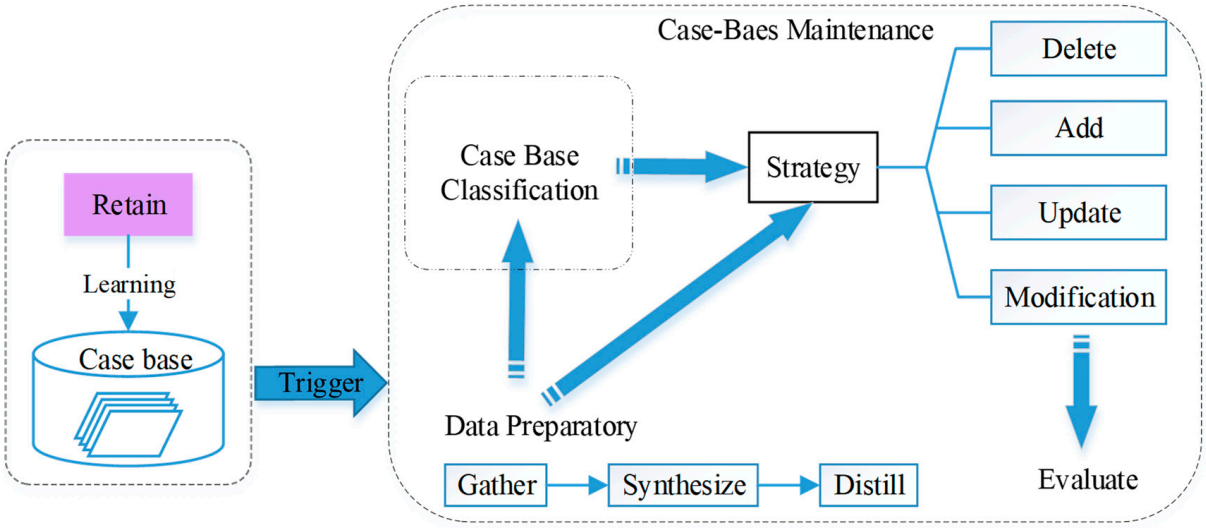


Figure 7. CBM maintenance process.

Although the direct model is simple, the effect is relatively poor. The case attribute model is maintained by adding additional information, but the selection and addition of additional information are limited by the case base itself, which needs to be analyzed on a case-by-case basis. Therefore, the hybrid model is currently the most popular research method. For example, Nakhjiri et al. [88] proposed a CBM model based on reputation values, calculating a case attribute called reputation for each member of the case base, whose value reflects the capability of the relevant case. Based on this case attribute, several removal strategies and maintenance methods are designed. Each strategy focuses on different aspects of case-based maintenance. Chebli et al. [95] proposed a CBM strategy based on active semi-supervised maintenance (ASSM), using ML technology to solve the problem of scarcity of marked cases. Smiti et al. [96] provided a new alternating technique to correctly detect noise and redundancy in cases or features. It is a method of dynamically maintaining the case base and can maintain the vocabulary knowledge carrier of CB and CBR systems at the same time.

4. Application Fields of CBR

Since the 1990s, CBR has been increasingly applied, including in diagnosis, decision-making, design and so on. To summarize the practical application of CBR, based on some keywords in the title and abstract of CBR literature, the author counted the number of literatures on several common applications. Specifically, it includes classification, diagnosis or diagnose, prediction or predict, design or plan, recommendation, decision making or decision support, and knowledge management. Among them, the literature search on planning and design was limited to the title. The trend of the number of application documents in different fields is shown in Figure 8. CBR applications are increasing and occupy an important position in the field of AI. In this chapter, some typical applications are mainly introduced from the following aspects: diagnosis, prediction, design and planning, decision support and recommendation systems. Moreover, other applications are briefly described.

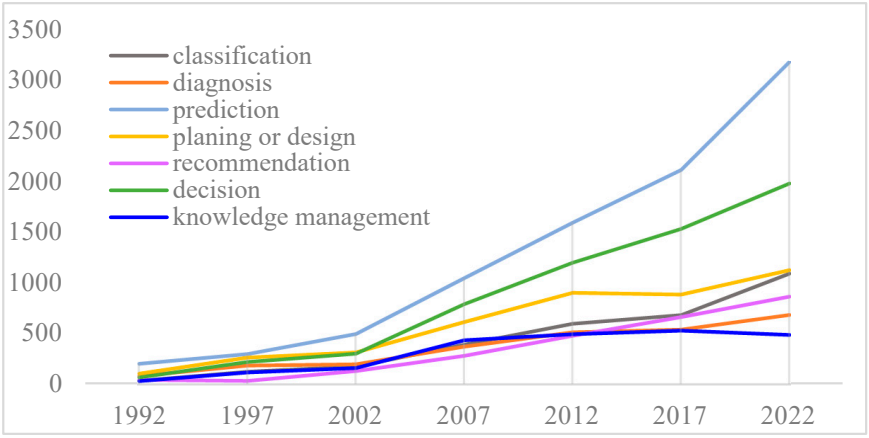


Figure 8. Variation trend of the number of literatures on different applications of CBR.

4.1. Diagnosis

Diagnostic tasks usually include technical fault diagnosis [97,98] in electronics, engineering and other fields and medical health case diagnosis [39,99,100], which can be regarded as a variant of classification tasks. Many AI methods can simply diagnose problems. However, CBR is different from these methods. The main advantage of using CBR to achieve the diagnosis task is not only to determine the diagnosis itself but also to provide a reference solution through CBR. The specific process is shown in Figure 9.

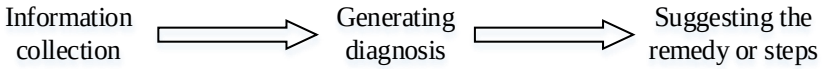


Figure 9. Diagnostic process.

With the development of technology, the function and structure of modern mechanical equipment or electronic systems have become increasingly complex, which makes the fault mechanism staggered and changeable. In addition, fault diagnoses of complex systems have become very difficult. Therefore, the fault diagnosis method based on CBR is mostly combined with other methods to improve the diagnosis efficiency. Chen et al. [98] developed an aero-engine fault system based on CBR. They designed a tree structure based on a semantic graph to quantify the similarity between semantic attributes and defined the relationship between fault components and fault modes to accurately match them.

Because the decision-making of the human medical health process itself is highly dependent on historical experience therapy and literature [39], medical pathology diagnosis based on CBR has always been a popular research topic in the medical field. Benamina et al. [99] established a fuzzy CBR application system on the JColibri platform to diagnose diabetes. They used symbolic learning to build a fuzzy decision tree on the Fispro platform, extract fuzzy rules, and import them into the JColibri platform to establish a fuzzy case base. The case index of the diabetes surveillance plan was selected by the retrieval method based on the fuzzy reasoning mechanism. This method not only optimizes the time but also reduces the complexity of the similarity calculation between individuals.

4.2. Prediction

In CBR, the application of prediction is very common, and the form of the case becomes:

$$case = \{history, predctioin\} \tag{5}$$

It is based on the fact that the history in the same class is similar in some sense, so the predicted results are similar. CBR is widely used in prediction tasks. In the industrial control process, the prediction of key parameters by CBR can improve the efficiency of industrial production, realize the stable control operation of the working condition process,

and reduce the safety risk, such as [101,102]. In financial forecasting, the analysis and prediction of financial data such as stocks can avoid risks and reduce the economic losses of shareholders or companies, such as [103]. In disaster relief emergency, CBR can realize the prediction of material demand with the help of previous disaster cases, which is the basic condition of disaster relief operation and the premise of optimal allocation of emergency resources, such as [104]. In healthcare, the prediction of body data by CBR can help people improve their physical fitness and reduce the risk of illness or injury, such as [105]. In agriculture, the prediction of yield by CBR can help farmers improve and optimize crop planting and increase yield, thus promoting the economic development of agricultural products, such as [106].

4.3. Design and Planning

Case-based design (CBD) [107–109] is a CBR-based application that develops new design solutions by adjusting previous solutions [30], allowing designs to be re-used and adapted to create new designs. For example, Grace et al. [107] proposed a dual-cycle CBR model in the domain of recipe generation. The model combines the strengths of deep learning and similarity-based retrieval. The model learns the latent structure of the domain from the case base to generate a new object that deliberately flouts that structure and then reintegrates the new object with known cases through adaptation to generate novel and valuable recipes. Ke et al. [109] proposed an intelligent design for remanufacture (DFREM) method based on a vector space model (VSM) and CBR, which can extract the features of customer requirement data from massive customer requirement data and quickly generate a design scheme meeting customer requirements.

The planning problem relies on the experience of decision-makers to propose a reasonable planning scheme. As one of the important applications of CBR, planning tasks [5,110–112], which are similar to design tasks, are not planned from scratch. For example, Abdelwahed et al. [112] proposed using learning experience to solve motion planning problems and CBR technology to provide state samples associated with the problem to achieve path building and find the shortest path. Chen et al. [5] proposed a CBR framework for assembly sequence planning (ASP). By establishing the overall architecture of CBR, the extraction, aggregation and re-use of assembly information of existing solutions are achieved, which can analyze the collection information in heterogeneous multimedia and re-use the existing assembly information scattered in multimedia solutions to solve assembly sequence planning.

4.4. Decision Support

The decision support system (DSS), based on CBR, uses data, knowledge, and reasoning technology to assist decision-makers, which can improve the level and quality of decision making, such as [113,114]. The decision-making process is roughly shown in Figure 10. The CBR system can provide reference and help for decision-makers in emergency responses, management, clinical situations, and other fields, with the help of historical cases that have occurred in the past.

In emergency, CBR can design emergency-management programs or auxiliary strategies to help managers make correct decisions in emergency affairs, in a timely manner, to avoid greater losses [4,113]. In management, CBR often provides managers or decision-makers with past management experience and helps them choose or specify the appropriate strategic guidance through the analysis of the management system, long-term development direction, and other aspects [115,116]. In clinical contexts, CBR provides reference information for experts through medical cases and helps experts analyze the patient's physical condition to realize the expert's determination of the patient's treatment plan [117,118].

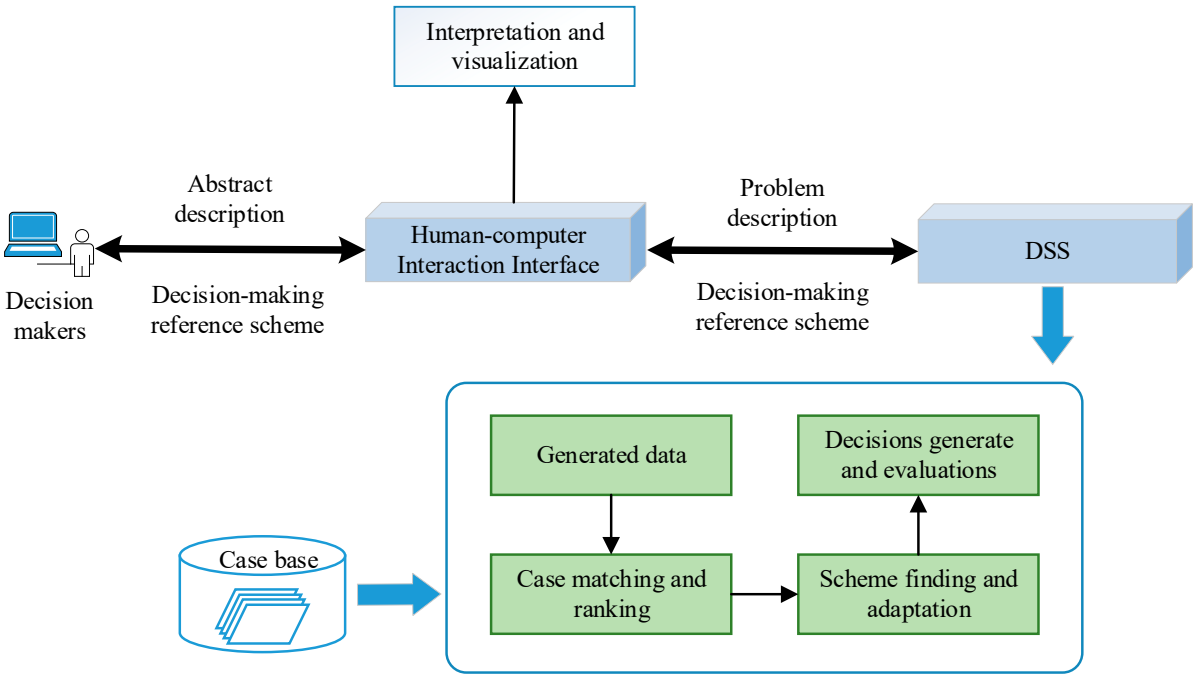


Figure 10. Decision-making process of the CBR system.

4.5. Recommendation System

E-commerce is similar to any other form of business. Users have needs, but often the needs can only be met to a certain extent, and users can only specify the needs incompletely or inaccurately. For users, it is difficult to analyze massive volumes of information and filter out the information they need; therefore, CBR plays a key role in the development of content-based or case-based recommendation systems, such as [119,120]. It includes the user's personal preferences and needs, as well as product-related information. Therefore, CBR can provide products and information that users are interested in or need according to user needs or preferences.

Abbas et al. [58] proposed a dynamic review recommender 'DiversityBite' based on the CBR framework, which combines reviews and diversity to help users obtain diversified recipes and meal plan recommendations. Dong and Smyth [121] proposed a personalized recommendation method using product reviews as recommendation knowledge. It extracts cluster features from reviews to form cases through triplet features and feature clustering steps. In the recommendation process, it calculates the recommendation score through the linear combination of emotional features and similarity, which is used for query-based and user-based recommendation scenarios. Li et al. [122] argued that, since users only interact with items of interest in the recommendation system, that system must retain very large amounts of personalized information and item sparsity, which seriously affects the performance of the recommendation system; therefore, they proposed a CBR-recommender method to reduce data sparsity through data classification and dynamic clustering, which makes the system run faster in large-scale recommendation research that dynamically calculates user preferences.

4.6. Other Applications

In addition to the above applications, CBR has also been applied in other fields, such as knowledge management (KM), and text and image processing.

KM is usually implemented through the knowledge flow of tasks such as creating, distributing, and re-using knowledge. The KM application program based on CBR is achieved by the knowledge base, which turns the knowledge base into a case base and achieves knowledge management with the CBR cycle process [55,123].

When the knowledge source of CBR is in text form, the method used is called text-based case-based reasoning; for example, the CBR method is applied to the Text-to-SQL task in semantic parsing [124], or the Date-to-Text generation task [125,126].

Image processing is an important research field in pattern recognition. Images can be used as query objects and can also appear in the solution to the problem. In other words, the image itself is an expression of knowledge. The description, interpretation, and retrieval of images are also important components in CBR applications. At present, the CBR method is mainly used as an auxiliary means of image processing to improve performance [127–129].

5. Summary and Challenge

Aha [1] and De Mantaras et al. [15] pointed out that the rise and development of CBR is influenced by cognitive science, analogical reasoning, knowledge-based systems, and uncertainty systems. Thus, this method has the advantage of being interdisciplinary. Moreover, the positive aspects of these factors can be used to improve reasoning. As shown by the above key technologies and application development, the CBR method also has some defects; for example, in the processing of unstructured data, CBR applications are relatively rare. In terms of the accurate prediction of parameters, it only provides approximate answers and cannot be applied to unsupervised learning problems. As Watson stated, CBR, as a methodology, can be a flexible combination of various technologies to achieve problem solving, interpretation, and learning processes, and can be applied to different research areas [10]. In recent years, AI technology represented by DL has made breakthrough progress and has attracted wide attention around the world. The research and application of CBR are also developing. Combined with the development needs of explainable AI (XAI) [130], according to the idea of DL, we believe that CBR may have the following two research directions in future development:

(1) Explainable CBR (XCBR) Decision Support

With the great success of methods such as ML and DL in AI, explanation has been identified as a key factor in the adoption of AI systems in a wide range of environments [131]. The XAI study attempts to address several issues related to the growing need for explanation models, such as; how do we design explanation models? How do we assess the resulting explanation? What knowledge do we need to construct explanations? How do we tell AI systems to provide users with suggestions and decisions, etc.? As mentioned at the beginning of this section, CBR can achieve the function of problem explanation and can be applied well in explanation tasks. It has been studied and applied in interactive interpretation and memory-based technology to generate interpretation, such as expert systems and recommendation systems. Therefore, CBR has significant potential and foundation in explaining opaque (i.e., black box) AI methods and establishing interpretable AI systems.

In 2022, the theme of the 4th XCBR workshop in the 30th ICCBR was case-based reasoning for the explanation of intelligent systems under the aim of XAI. At present, CBR has made many contributions to XAI research, such as image explanation tasks [132], text explanation tasks [133], and neural network-based explanation tasks [134]. However, CBR is only moderately used in XAI or in certain aspects of intelligent systems; therefore, regarding the interpretability of complex AI systems, CBR still has much room for development, such as [135,136].

On the one hand, the interpretation application of CBR in AI systems is mainly oriented to users and provides decision support for users with interactive interpretation; however, in fact, the object-oriented aspect of a complex system is not only the users, and should also include the system operation and maintenance of staff, technical personnel, and other related personnel. Therefore, the interpretation should vary from person to person, using CBR to achieve diverse interpretation. How CBR achieves different interpretations according to different objects in AI systems will be an interesting and challenging research direction. The diversity of object-oriented interpretation means that XCBR needs to include the interpretation cases of different objects in the case base. This is not a simple case list, and its construction problem needs to be solved. At the same time, when new objects and

new cases appear, it means the emergence of new explanations, which update and maintain the case base.

On the other hand, how to establish a user-centered personalized interpretation model has always been a problem worthy of study. Darias proposed that the CBR system is a living system, and XAI is a constantly changing research field. The case base must update new interpreters to adapt to new problems and solutions; however, one of the biggest shortcomings of the existing XAI library is the lack of personalized interpretation [137]. Darias et al. [65] made attempts to address these shortcomings, and they have mentioned that this is only the first step in personalized interpretation models. Therefore, it is a long-term development outlook to provide personalized interpretation according to different user intentions and needs. It involves multi-domain content, user data representation, AI black box models, and demand analysis, etc. Once these problems and challenges are solved, it is believed that XCBR will go further in the field of AI.

(2) Deep CBR

In recent years, great success in the field of DL has prompted the CBR community to commit to applying DL technology to CBR. CBR, as a methodology, has a complete set of reasoning processes. These processes always need the support of other algorithms and technologies, and DL technology has unparalleled learning ability for large and complex case data, undoubtedly providing new research directions and technical support for CBR research. Table 3 describes the differences and similarities between the two from data, results, learning methods, and other aspects. David proposed that the challenge for the CBR community is to define necessary tasks and integration. Introducing DL components into a CBR system in order to realize CBR function can reduce knowledge burden and improve flexibility, and points to the problems faced by its application in case representation, adaptation, and so on. At the same time, David also studied the influence of network architecture on the quality of CBR feature extraction [138], which further promoted the development of DL and CBR.

Table 3. Deep learning versus CBR.

Deep Learning	CBR
Data, experience, and knowledge are all examples	Cases
It is about learning knowledge	It is about learning knowledge
General rules and laws are generated	Specific solutions are generated
Technology	Methodology
Unsupervised learning possible	Unsupervised problem solving cannot be done
Eager learners	Lazy learners
Results are not precise or certain	Results are not precise or certain

As mentioned in the description above, DL technology has been applied in various steps in CBR systems [74,118,130]; however, the application of the DL method in CBR is scattered, which is only reflected in a certain step or in some simple combination with the CBR method. It is a direct application based on representation; that is, only some algorithms in DL are selected to implement a process in CBR. Although this can improve the performance and efficiency of CBR systems to a certain extent, it is only the beginning of deep CBR research. It lacks a more abstract general CBR system based on DL idea, or a high-level framework process to integrate advanced CBR with DL architecture. Leake and Crandall [30] advocated using CBR to promote the development and challenges of DL. Can we, in turn, use DL thinking to design new CBR frameworks and methods to achieve the whole process of reasoning, learning, and explanation? Alternatively, how can we realize deep reasoning in CBR? Here, depth refers not only to the comprehensive application of DL technology in CBR, but also to deeper learning and reasoning based on the CBR structure.

Researchers at Deakin University proposed the implementation from DL to deep reasoning at the 2021 International Conference on Knowledge Discovery and Data Mining (KDD). The emphasis in this tutorial is still on implementing reasoning in DL. Obviously,

CBR is more advantageous at the reasoning level, and its lazy learning can reduce the burden of learning to a certain extent. Therefore, further consideration of the combination of CBR and DL in applying deep CBR to complex AI systems may be a future development direction in CBR. To achieve this, a bridge between the two methods needs to be established. As Glatt et al. [83] have noted, there are relatively few studies on the CBR community considering reinforcement learning (RL) methods to accelerate CBR. This lack of research may be because the differences in formula and vocabulary hinder the exchange of ideas and methods between communities; therefore, CBR systems based on DL also need to consider how to use DL language or symbols to describe the knowledge in CBR, such as vocabulary knowledge, case knowledge, adaptation knowledge, or rules. This is the first step in deep CBR, reflecting improvements in methodology and technology.

After the first step of combining DL and CBR, we should further consider the concept of 'deep'. As deep learning realizes the depth of learning by the depth of the network, deep CBR should realize the depth of reasoning with the depth. To date, CBR is limited to a 4R cycle reasoning process. Can CBR be explored in depth, based on this process? For example, can deep CBR be achieved by the nesting of multiple cycles? It is not necessarily achieved through multiple cycles of a complete 4R cycle, but, rather, through a process of several alternating steps in the 4R cycle. Alternatively, the 4R cycle process can be extended and refined into more detailed reasoning steps to achieve deep reasoning. This is an improvement upon the deep CBR structure. If these ideas and technical difficulties are realized, it will be possible to promote the development of CBR to new heights.

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